



Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams

<https://fani-lab.github.io/OpeNTF/tutorial/recsys26/>



fani-lab.github.io



recsys.acm.org/recsys26/



TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

NEURAL TEAM RECOMMENDATION

TIME AND LOCATION

Half-day, Afternoon, 1:30 PM - 5:30 PM (GMT/UTC -6), Monday, September 28, 2026

Elk/Deer Lake (4th floor), Marriott City Center, Downtown, Minneapolis, Minnesota, USA

Prev Tutorials at UMAP24, SIGIR-AP24, and WSDM25, and CIKM25 < [Full Outline] [Slides] [Recording]

[Course Home](#)[Content](#)[Classlist](#)[Grades](#)[Quick Eval](#)[Class Progress](#)[Course Tools ▾](#)[UVic Resources ▾](#)[Help ▾](#)[Overview](#)[Bookmarks](#)[Course Schedule](#)[Table of Contents](#)

29

[Project](#)

4

[Team Projects List](#)[1. Project](#)

1

[Definition and
Team Formation](#)[2. Background and
Related Work](#)

1

Team Projects List ▾

[Print](#)[Settings](#)

Add dates and restrictions...



A: Ali, R: Reia

Team ID	Student	Project Title	K
1A	Mason McNeill	Identifying Manipulative Twitter Bots Through Metadata Analysis	S
	Ken Wright		F
	Arunkumar Krishnamoorthy		T
	Carter Dunstan		M
10A	Douglas MacGregor	Image-Based Classification of British Columbia Seaweed	C
	Arda Berklin		S
	Jarren Morris		M
			In
1R	Alicia Kim	Understanding URL Characteristics Behind AI-Generated Phishing Evasion	K
	Jay Cheng		B
	Anson Koh		S
	Owen Pritchard		P
2R	Abhishek Majumdar	PRISM: Evaluating Privacy-Constrained Customer Churn Classification	M
	Dev Parsana		A
	Ravi Vashi		
	Bharath Irukulapati		
3R	Hung Vuong	MajorMatch: From Hobbies to Major	
	Brendan Wong		
	Robert Legassicke		
	Yashwanthkrishna Nagharaj		



Search Topics



Overview



Bookmarks



Course Schedule

Table of Contents

29



Project

4



Team Projects List

1. Project

1

Definition and
Team Formation2. Background and
Related Work

1

Team Projects List



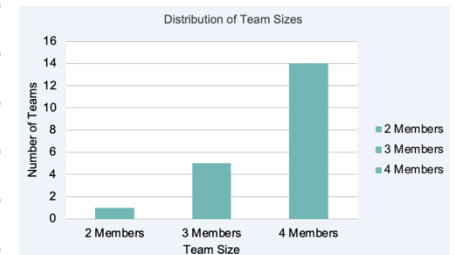
Print



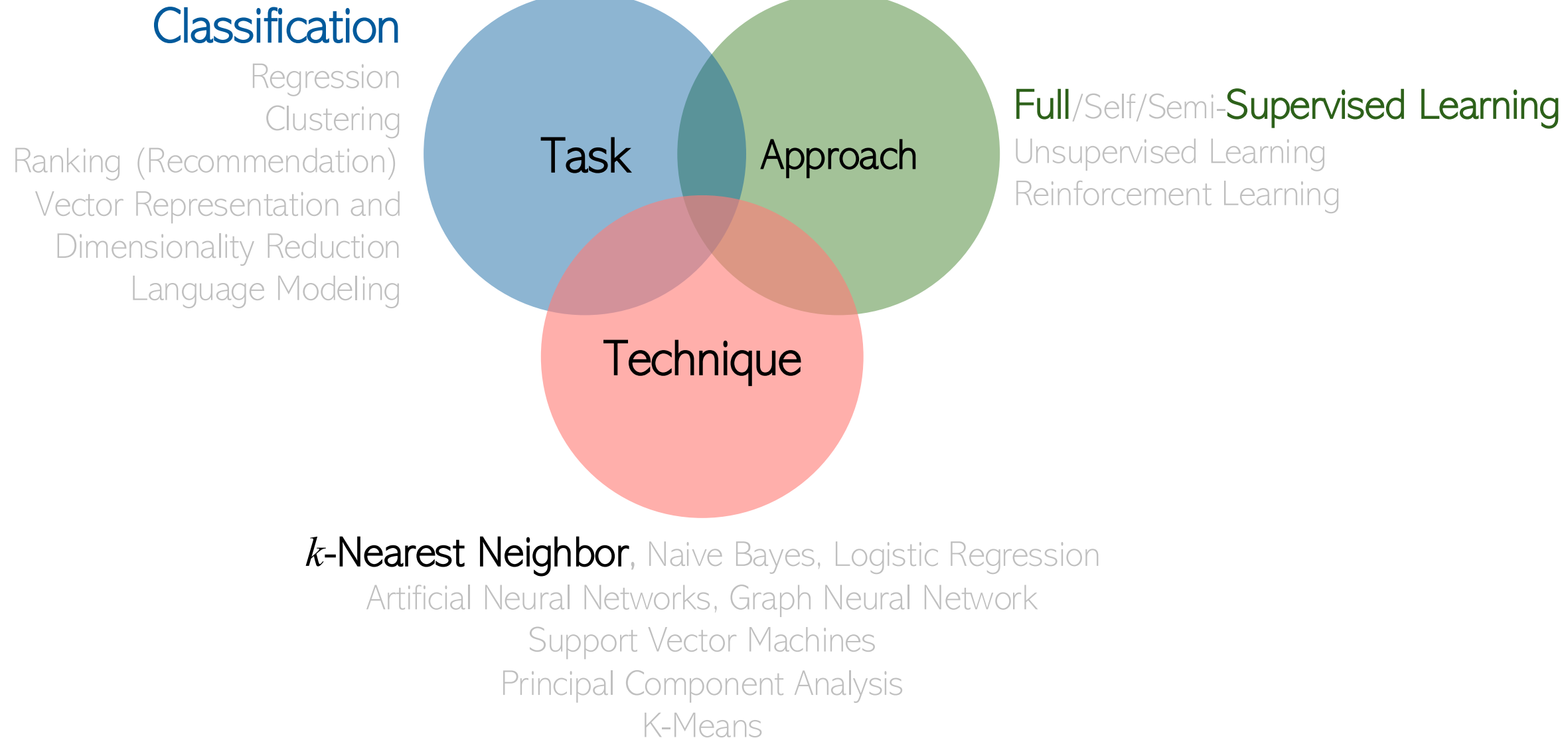
Settings

Team size	Number of teams	Students	Percentage of teams
2	1	2	5%
3	5	15	25%
4	14	56	70%
Total	20	73	100%

Theme	Projects	Share
NLP / Social Media / Text Mining	5	25%
Cybersecurity / Fraud / Phishing	4	20%
Environment / Climate / Natural Hazards	4	20%
Health / Biomedical	3	15%
Recommendation / Personalization	3	15%
RL / Agents	2	10%
Other / Networking / Audio	2	10%



4 Outline: W02B: Topics Start



5 Outline: W03A: Topics Start

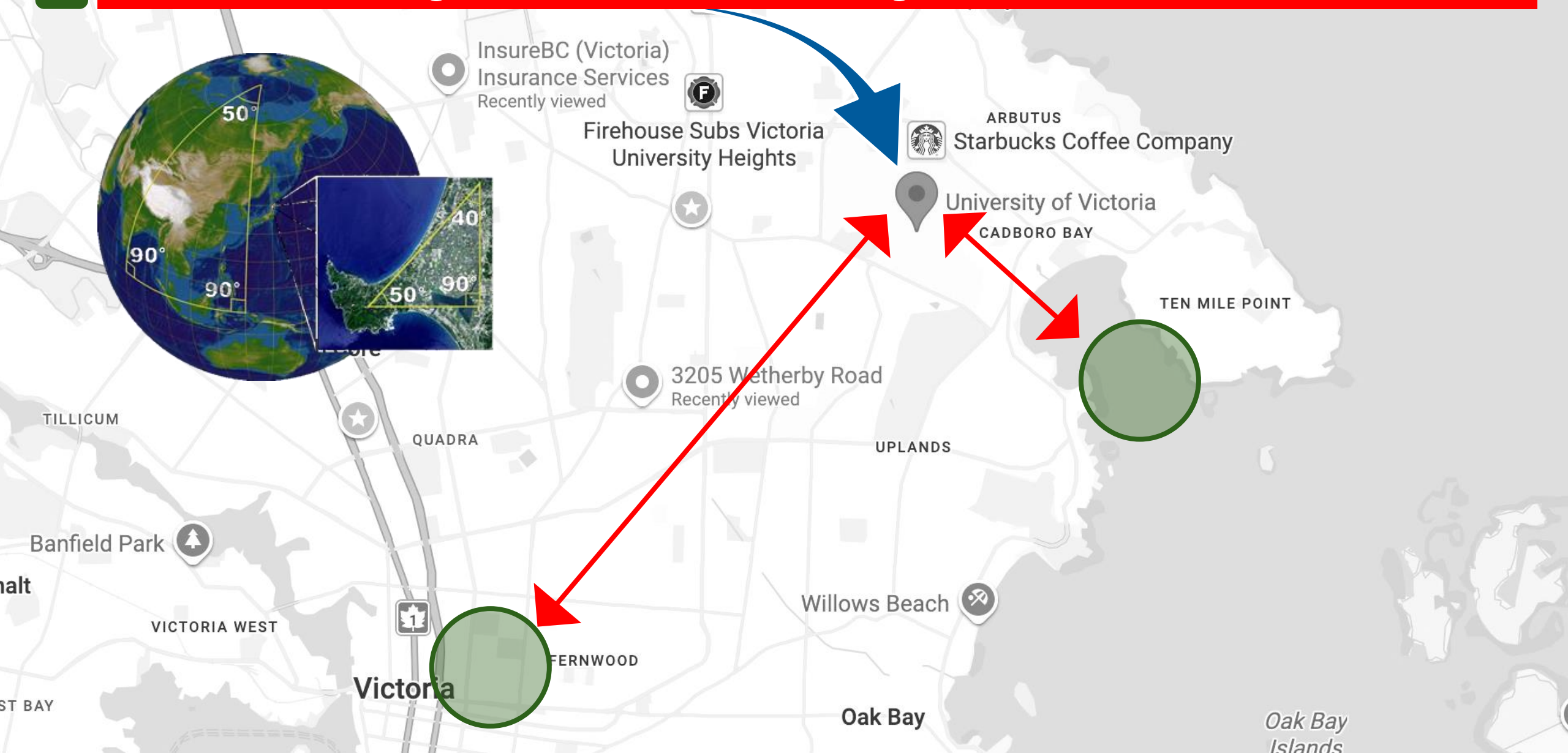
- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Let's zoom in ... Hyperparameters
- III) Evaluation

6

 k -Nearest Neighbor: Distance Function

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k -Nearest Neighbor: Math Challenges



8

k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.

More restrictions

More assumptions

More structure

More idealized

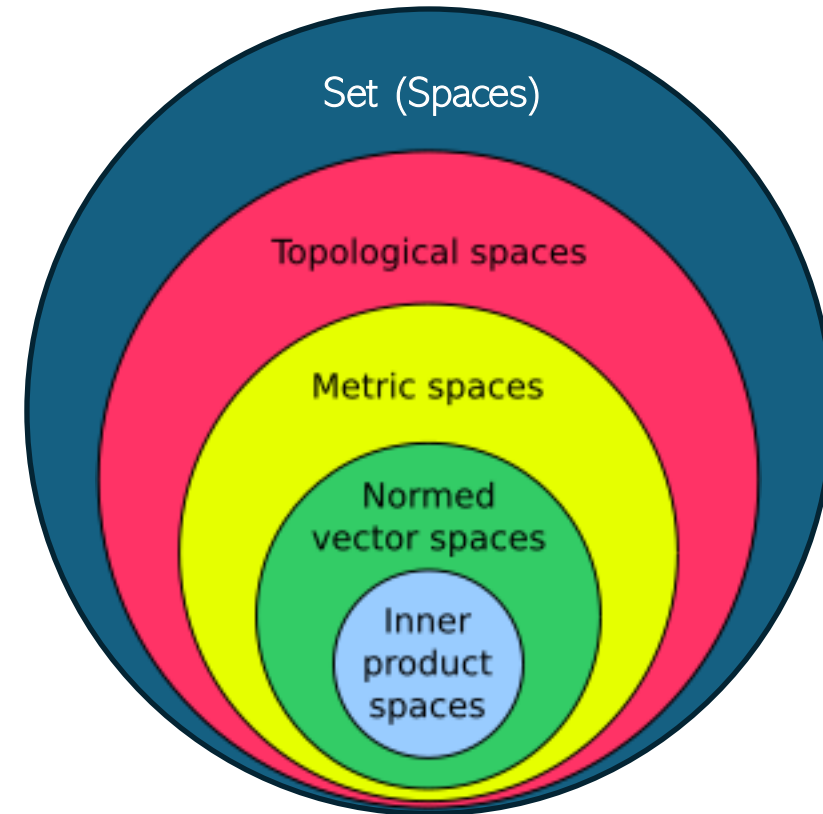
Less realized

Less general

Easier mathematics

Easier to work with

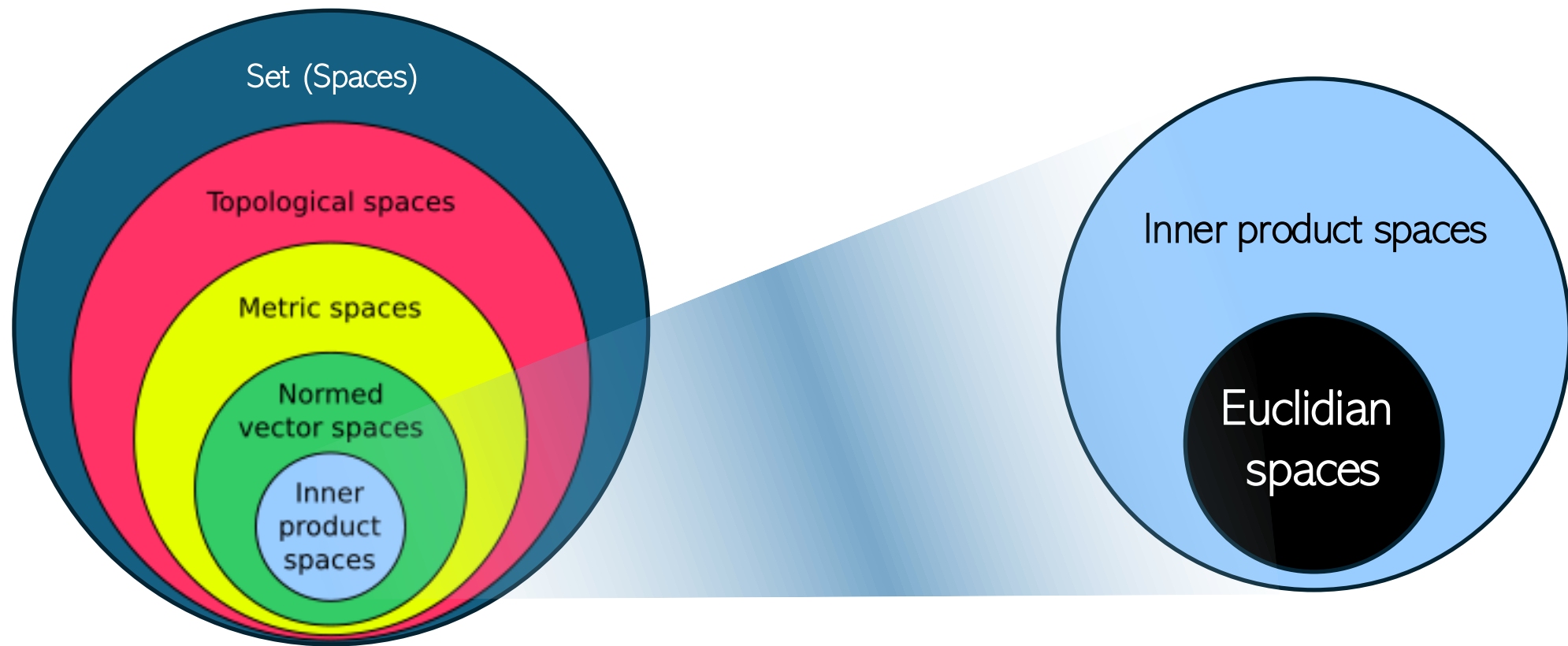
“give up generality to gain tractability”



9

k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.



“nice geometric space”

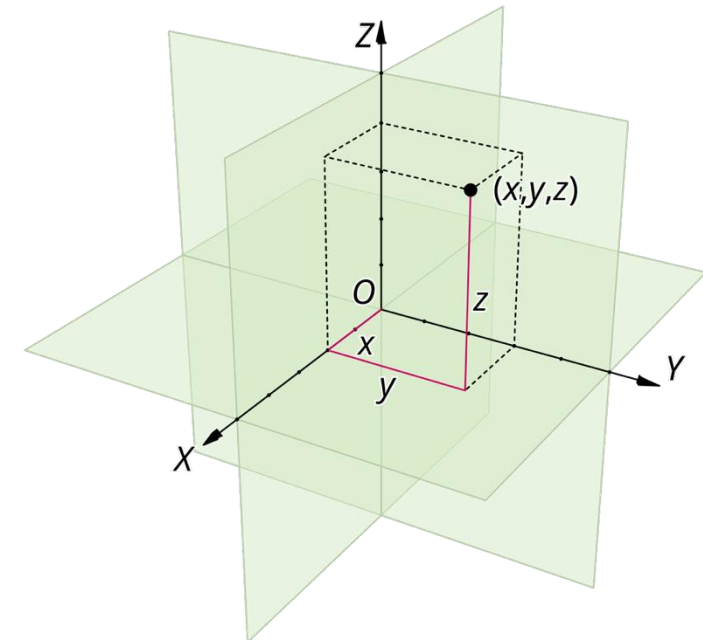
yoo-KLID-ee-uhn

10

k -Nearest Neighbor: Euclidian Space

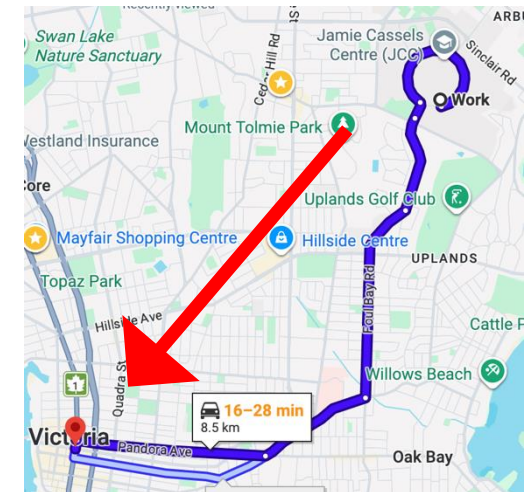
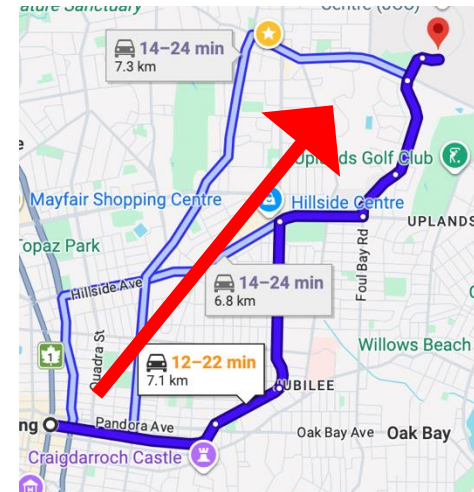
If $\mathbf{p}$ and $\mathbf{q}$ are two “points” (pls don't use the term “vector” for now):

1. $d(\mathbf{p}, \mathbf{q}) \geq 0.0$ (no negative distances).
2. $d(\mathbf{p}, \mathbf{q}) = 0.0 \Leftrightarrow_{\text{iff}} \mathbf{p} = \mathbf{q}$.
3. $d(\mathbf{p}, \mathbf{q}) = d(\mathbf{q}, \mathbf{p})$ (symmetric).
4. $d(\mathbf{p}, \mathbf{q}) \leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).



k -Nearest Neighbor: *non*-Euclidian Space

1. $d(\mathbf{p}, \mathbf{q}) \geq 0$ (no negative distances).
 2. $d(\mathbf{p}, \mathbf{q}) = 0 \iff_{\text{iff}} \mathbf{p} = \mathbf{q}$.
 3. $d(\mathbf{p}, \mathbf{q}) \neq d(\mathbf{q}, \mathbf{p})$ (symmetric).
 4. $d(\mathbf{p}, \mathbf{q}) \not\leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).
- $\geq$



k-Nearest Neighbor: Minkowski Distance



Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

$p = 1$; Manhattan Distance
 $p = 2$; Euclidean Distance
 $p = +\infty$; Chebyshev Distance

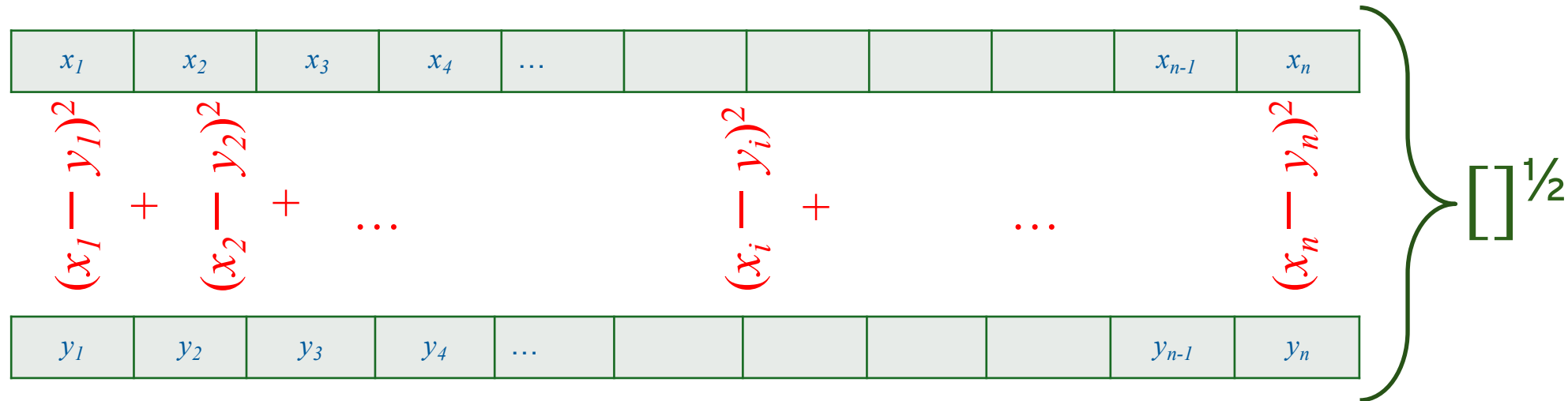
Hermann Minkowski (1864 – 1909)

Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen
Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905).

13

k -Nearest Neighbor: Curse of Dimensionality $\mathbb{R}^n \rightarrow +\infty$

$$\mathbb{R}^n \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{\frac{1}{2}}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

Each distance $\rightarrow n$ (subtractions + squares)

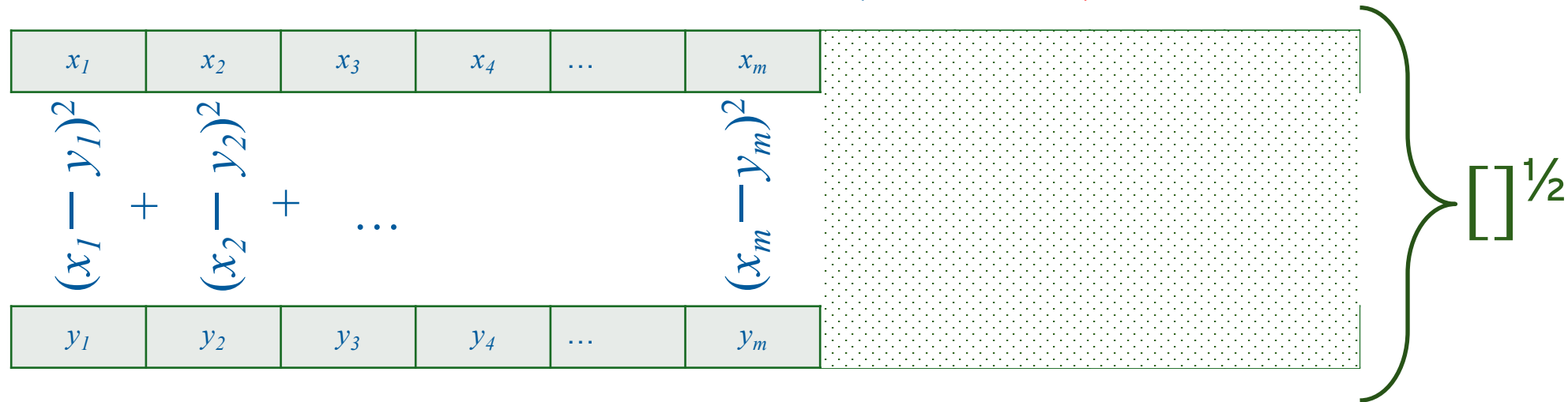
$\rightarrow n \times |\mathbf{O}^*| \rightarrow$ Linearly grows for both the size of training set and dimensions

14

k -Nearest Neighbor: Shadowing (Projection)

Naïve Dimensionality Reduction: Simply take the first few dimensions

$$\mathbb{R}^{n \gg 0} \rightarrow \mathbb{R}^{m > 0} \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, m, \dots, n\}} (x_i - y_i)^2]^{1/2}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

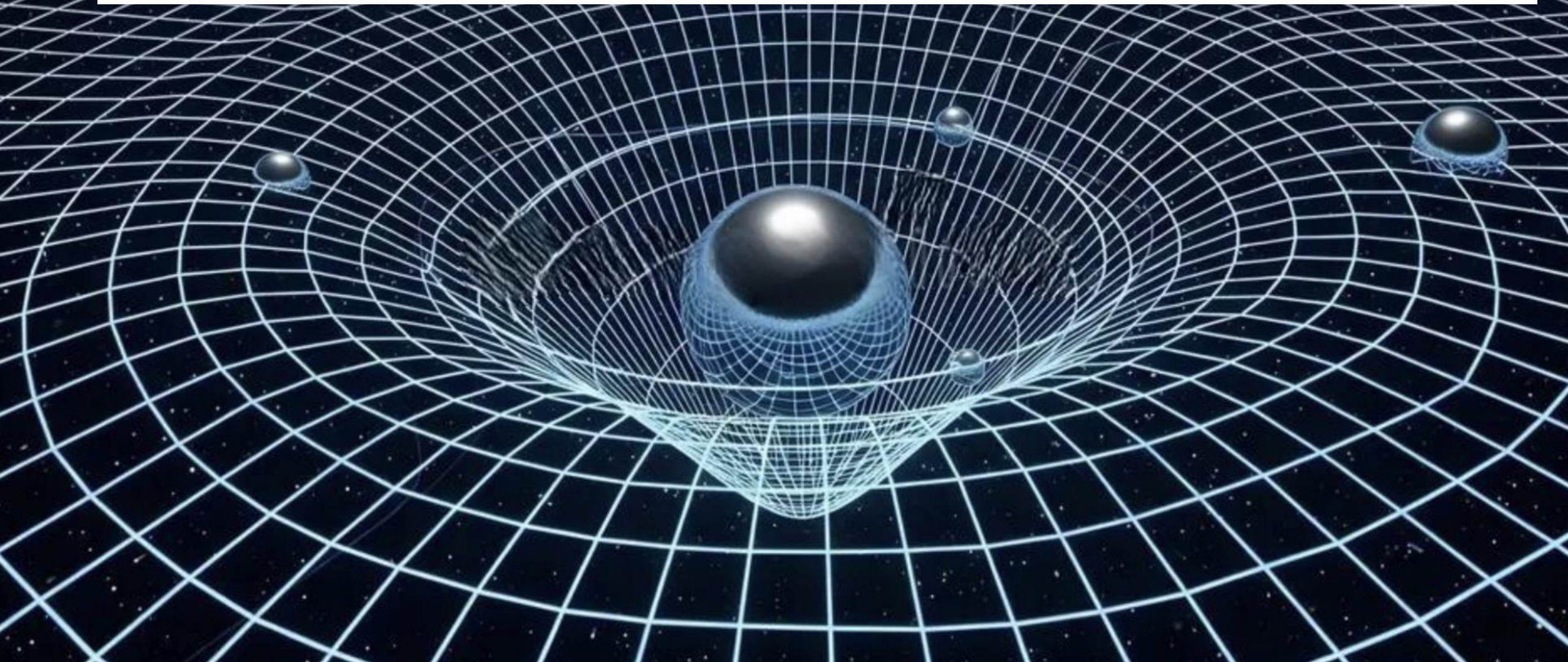
Each distance $\rightarrow m$ (subtractions + squares)

$\rightarrow m \times |\mathbf{O}^*| \rightarrow \text{Manageable}$

k -Nearest Neighbor: Locally-Sensitive Hashing (LSH)

Hashing Collisions for Efficient Nearest Neighbor Search → Almost Constant (1-3) Steps

CSC502: Systems for Massive Datasets, Mining of Massive Datasets: Ch03: Finding Similar Items



16

k -Nearest Neighbor: Domain Challenges

How to map a real-world observation to a point in Euclidian space?

People

Animals (Birds, ...)

Objects (Houses, Cars, ...)

Image (Biometric Iris)

Audio (Voice, Music)

Video (Movies)

Text (Words, Sentence, Docs)

Timeseries

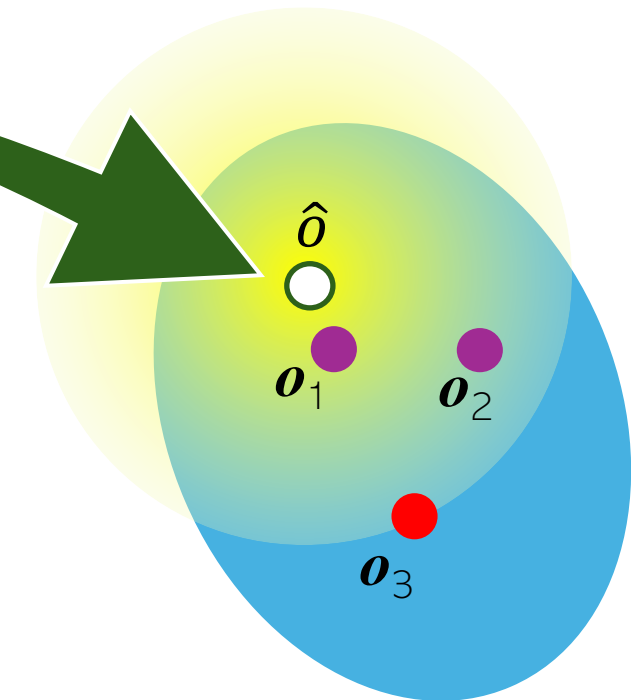
Genes

Spatial (GIS)

Light Bulbs

...

Encoding
Embedding
 $e(o)$



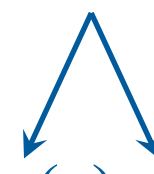
Euclidian Space
(Embedding Space)

17

 k -Nearest Neighbor: Domain Challenges

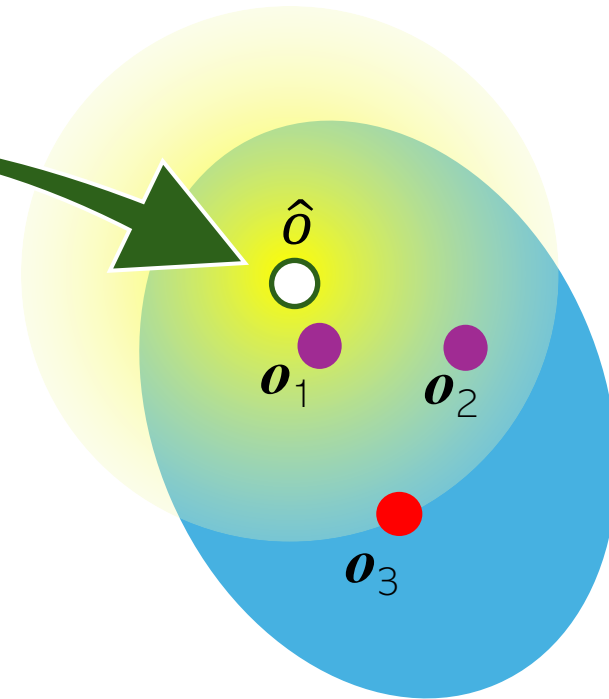
$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


$$d(\hat{o}, o) \approx d(e(\hat{o}) \in \mathbb{R}^n, e(o) \in \mathbb{R}^n) = d(\mathbf{x}_{\hat{o}}, \mathbf{y}_o) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{1/2}$$


$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(e(\hat{o}), e(o))\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


18 k -Nearest Neighbor: Text

Words
Sentences
Docs
Social Posts
Books
Webpages
...

 $e(o)$ 

Euclidian Space
(Embedding Space)

k -Nearest Neighbor: Text: Word Space $\rightarrow$ Euclidian Space

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C} = \mathbf{Entities} = \{\text{Person, Organization, Event, Food, Animal}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(w, c)\}$, $\mathbf{U}^* = \{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U} = \mathbf{Vocabs} \rightarrow \mathbb{R}^{n=1}$

20

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=1}$

Indexing $\rightarrow$ Enumerate words in a language's dictionary (vocabulary set)



$$e(\hat{w}: \text{'aardvark'})=0$$



$$e(w: \text{'apple'})=1050$$



$$e(w: \text{'zyzzyva'})=|V|$$

Does $e(.)$ preserve “real-world” proximity to “spatial” proximity? **No!**

$$\begin{aligned} |e(\text{'aardvark'}) - e(\text{'apple'})| &< |e(\text{'aardvark'}) - e(\text{'zyzzyva'})| \\ |0 - 1050| &< |0 - 520,000| \end{aligned}$$

'aardvark' is closer to 'apple' vs. 'zyzzyva'!

'aardvark' is tech company!

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C}=\mathbf{Entities}=\{\text{Person, Organization, Event, Date, Location}\}$

Approach: Fully Supervised $\mathbf{O}^*=\{(w, c)\}$, $\mathbf{U}^*=\{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U}=\mathbf{Vocabs} \rightarrow \mathbb{R}^{n=?}$

What is the “*real-world*” proximity in this problem?

→ Words’ semantics (meanings)

Two semantically close words are similar entities.

Two semantically distant words are dissimilar entities.

→ How to define meaning of a word?

22

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to **define** meaning of a word? Lexical approach by linguistics...

cat noun, often attributive

 Save Word

\ 'kat  \

Definition of *cat* (Entry 1 of 5)

New unknow/complex words

Polysemy

- 1 **a** : a carnivorous mammal (*Felis catus*) long domesticated as a pet and for catching rats and mice
- b** : any of a family (Felidae) of carnivorous usually solitary and nocturnal mammals (such as the domestic cat, lion, tiger, leopard, jaguar, cougar, wildcat, lynx, and cheetah)
- 2 **a** : GUY
// some young ... *cat* asked me to go drinking with him
— Jack Kerouac
- b** : a player or devotee of jazz
- 3 : a strong tackle used to hoist an anchor to the cathead of a ship

23

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to **define** meaning of a word? **We cannot!** E.g., “*game*”

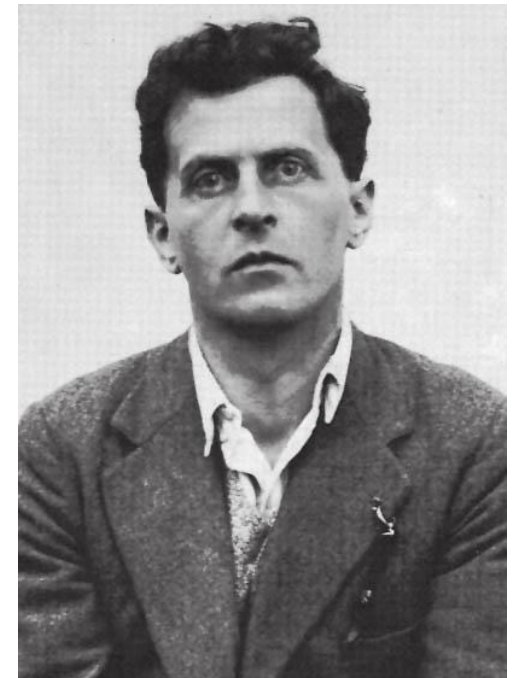
Skeptical of a formal theory of meaning definitions for each word

“The meaning of a word is its use in the language” - Philosophical Investigations.

Ludwig Josef Johann Wittgenstein
/'**ɪ**tgənʃtaɪn, -staɪn/
1889 –1951

Austrian-British Philosopher

Worked in logic, philosophy of mathematics, philosophy of mind, philosophy of language



24

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to ~~define~~ **capture** meaning of a word? **Distributional Hypothesis**

H_1 : Words occur in similar contexts tend to have similar meanings.

Like k -NN idea of birds of a feather flock together, but

It may or may not be correct!

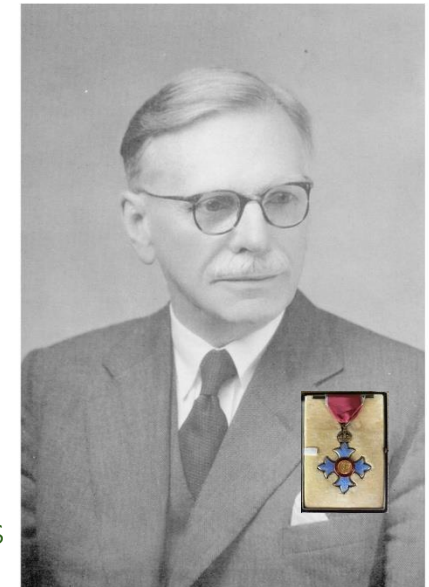
Joos, M. (1950). Description of language design.

Harris, Z. S. (1954). Distributional structure.

Firth, J. R. (1957). A synopsis of linguistic theory.

John Rupert Firth OBE (1890 – 1960)

Leading figure in British linguistics



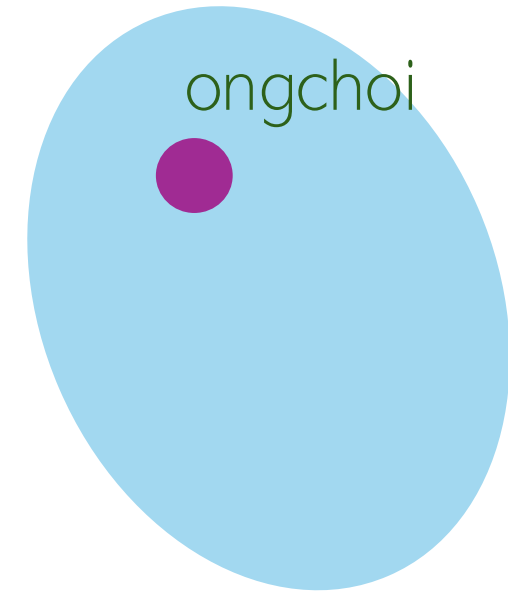
JOHN RUPERT FIRTH

25

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H_1 : Distributional Hypothesis

“ongchoi delicious sauteed garlic ...”



Euclidian Space
(Embedding Space)

26

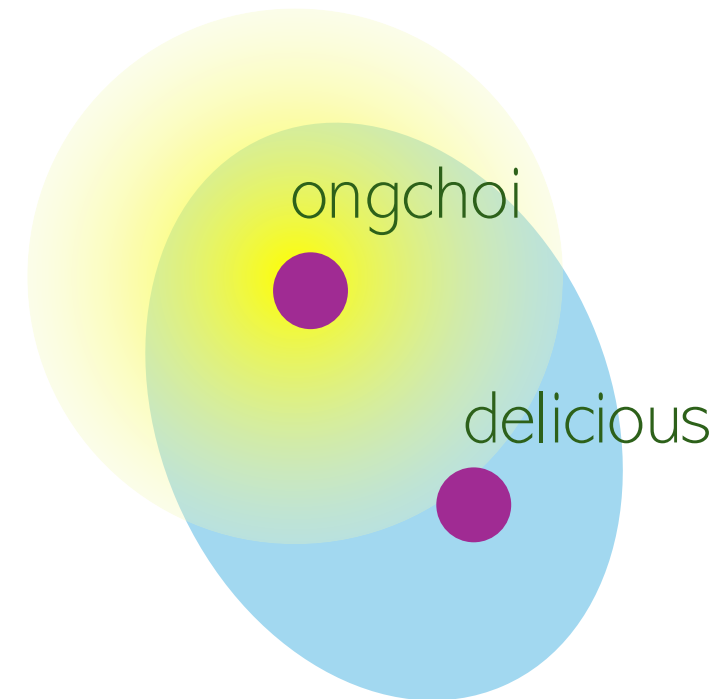
k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

“*ongchoi delicious sauteed garlic ...*”

context = 2 words

(ongchoi, *delicious*) = 1



27

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

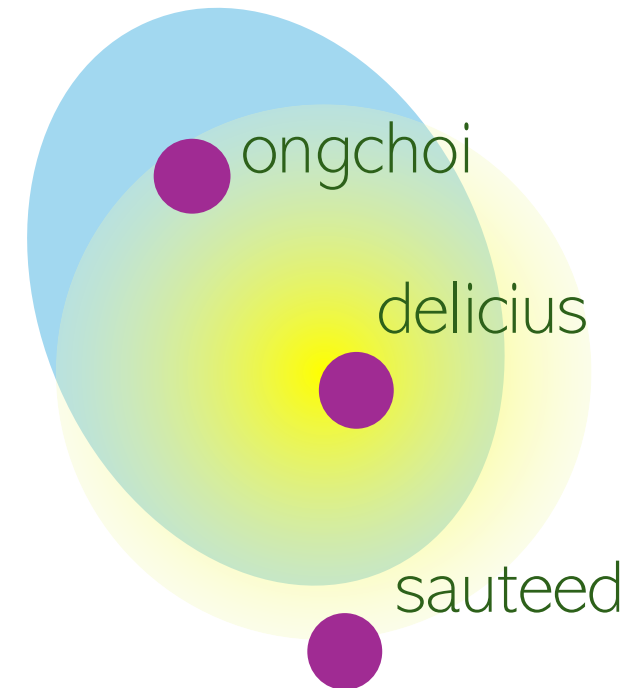
How to capture meaning of a word? H1: Distributional Hypothesis

“ongchoi delicious sauteed garlic ...”

context = 2 words

(ongchoi, delicious) = 1

(delicious, sauteed) = 1



28

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

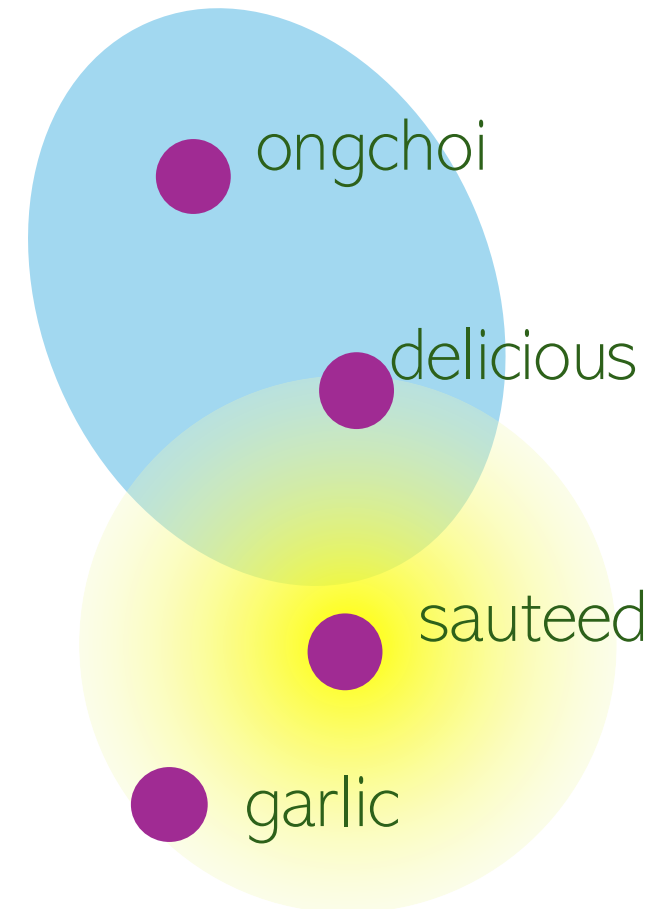
“ongchoi delicious sauteed garlic ...”

context = 2 words

(ongchoi, delicious) = 1

(delicious, sauteed) = 1

(sauteed, garlic) = 1



29

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

“ongchoi delicious sauteed garlic ...”

“sauteed ongchoi ...”

context = 2 words

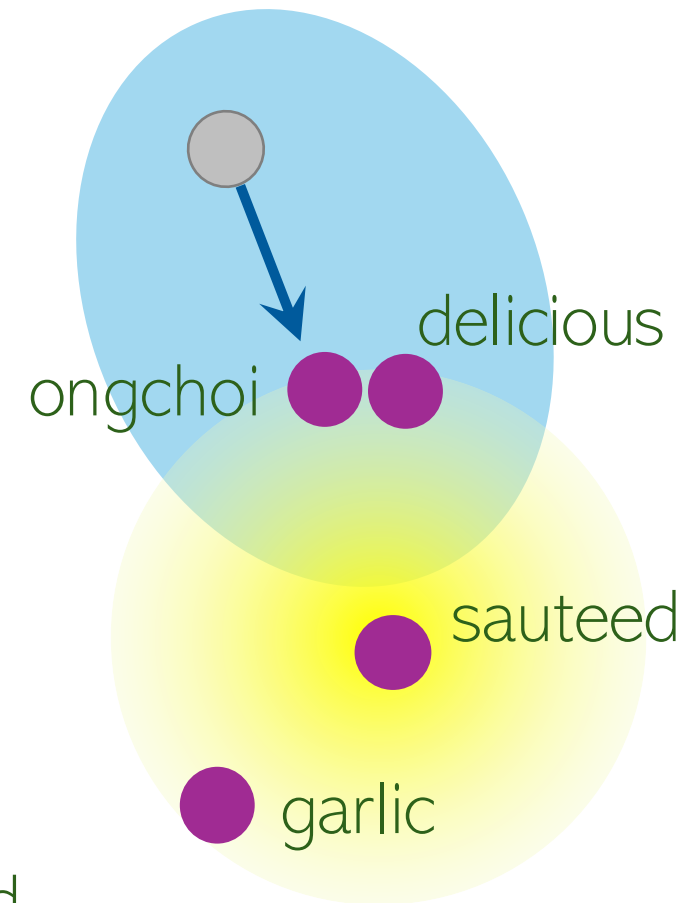
(ongchoi, delicious) = 1

(delicious, sauteed) = 1

(sauteed, garlic) = 1

(sauteed, ongchoi) = 1

Ongchoi is not Person, nor Organization, but Food.



30

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

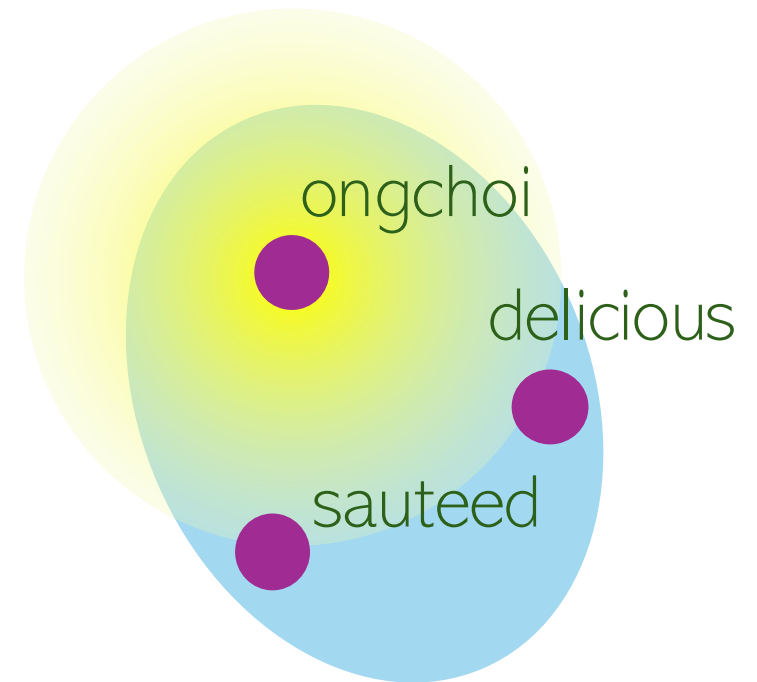
“ongchoi delicious sauteed garlic ...”

context = 3 words

(ongchoi, delicious) = 1

(delicious, sauteed) = 1

(ongchoi, sauteed) = 1



31

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

“ongchoi delicious sauteed garlic ...”

context = 3 words

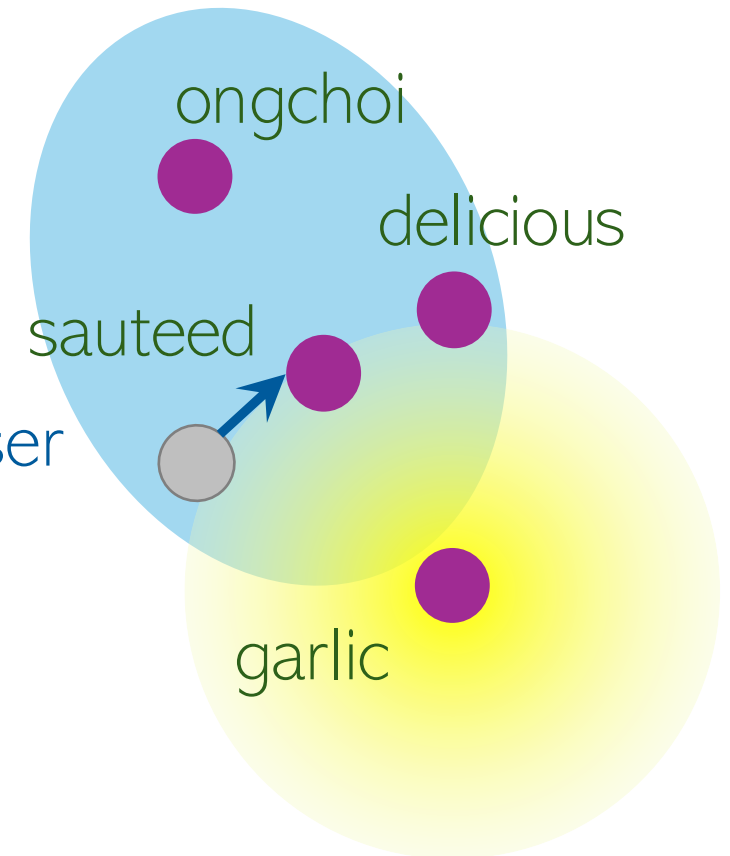
(ongchoi, delicious) = 1

(delicious, sauteed) = 1 + 1, more cooccurrence, closer

(ongchoi, sauteed) = 1

(delicious, garlic) = 1

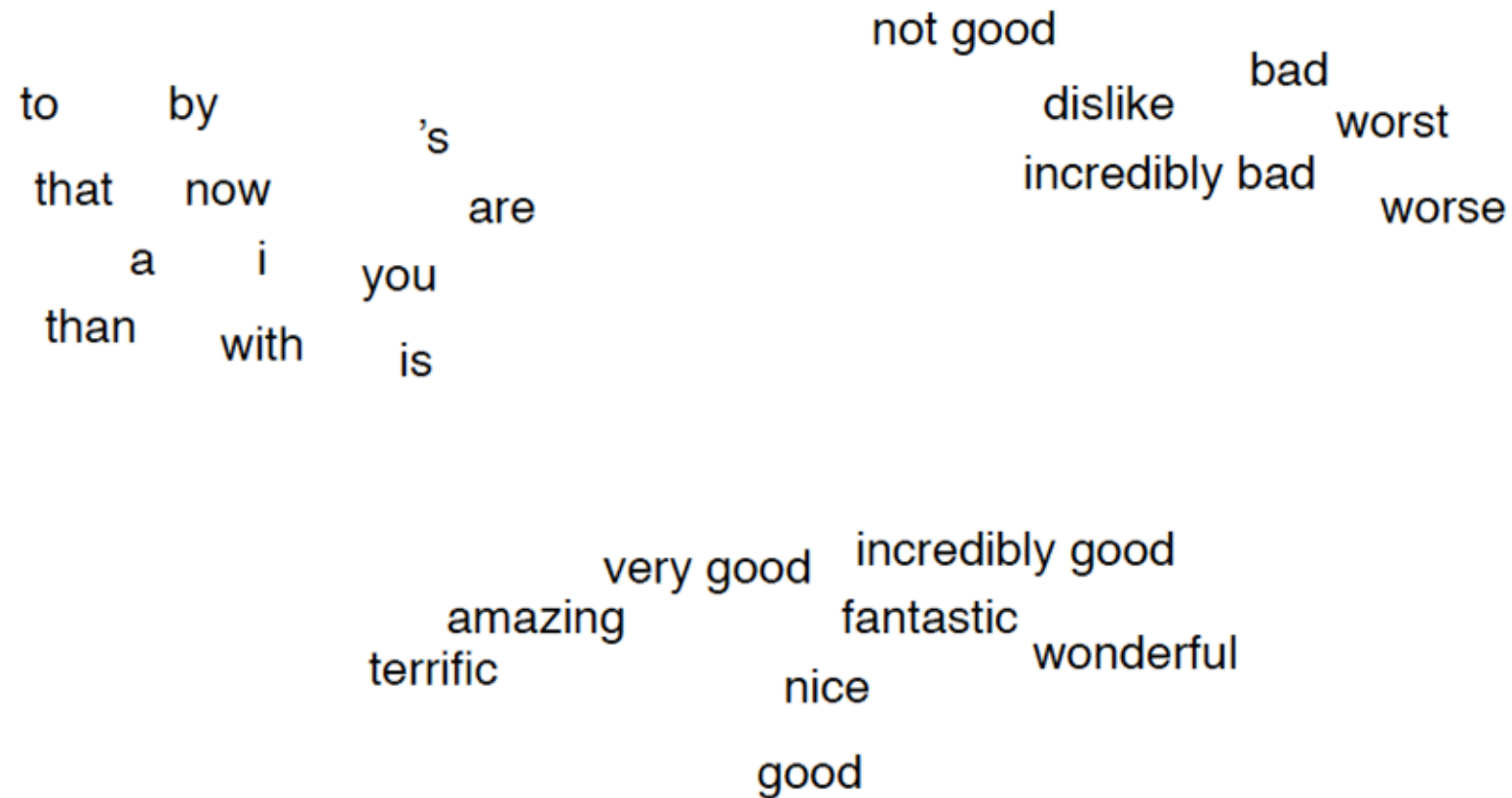
(sauteed, garlic) = 1



32

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis



Our mind if 2D Euclidian.

33

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: **Distributional Hypothesis**

Location of the words does *not* matter.

More cooccur in the same context

- More similar semantic
- Closer meaning
- Closer the distance!

Linguistic Semantic (Real-World) Proximity

- Cooccurrence Context in H1
- **Distributional Semantics**
- Geometric Proximity

Computational Leverage

not good
dislike
bad
worst
incredibly bad
worse

amazing
terrific
very good
incredibly good
fantastic
wonderful
nice
good

34

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

~

$\rightarrow$ Cooccurrence

Semantic Transparency

The degree to which you can guess the meaning of a word, compound, or idiom by its surface. e.g., “eschew obfuscation”, “Zyzzzyva”, ...

[https://en.wikipedia.org/wiki/Transparency_\(linguistic\)](https://en.wikipedia.org/wiki/Transparency_(linguistic))

Note it down, there is an exam question about this ...